

ACP

AARAMBH CHAPTERWISE PROBLEMS

Class-9th

CROSS >>> POLICE LINE >>>

E >>> DO NOT CROSS >>> LINE >>>



GRAVITATION

PRASHANT KIRAD



GRAVITATION

↻ (Options acche se padhna)

MULTIPLE CHOICE QUESTIONS (1 Mark)

1. The motion of the moon around the earth is due to:
(a) Centripetal Force
(b) Electrostatic Force
(c) Magnetic Force
(d) Centrifugal Force
2. A body has a mass of 2 kg. When will the mass of the body change?
(a) When the body is taken to the moon.
(b) When the body is dropped from a height.
(c) When the body is being pulled along a smooth surface.
(d) The mass of the body will not change unless it is cut or broken.
3. A girl stands on a box having dimensions 60 cm × 40 cm × 20 cm in different orientations. In which case will the pressure exerted on the ground be maximum?
(a) maximum when length and breadth form the base
(b) maximum when breadth and width form the base
(c) maximum when width and length form the base
(d) the same in all the above three cases
4. An object is placed in three liquids of different densities. It floats with $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{3}{4}$ of its volume submerged in liquids with densities d_1 , d_2 , and d_3 respectively. Which is correct?
(a) $d_1 > d_2 > d_3$
(b) $d_1 < d_2 > d_3$
(c) $d_1 < d_2 < d_3$
(d) $d_1 > d_2 < d_3$
5. The value of acceleration due to gravity 'g':

- (a) is same on equator and poles
- (b) is least on poles
- (c) is least on equator
- (d) increases from pole to equator

6. An apple falls from a tree because of gravitational attraction between the earth and apple. If F_1 is the magnitude of force exerted by the earth on the apple and F_2 is the magnitude of force exerted by apple on earth, then

- (a) F_1 is very much greater than F_2
- (b) F_2 is very much greater than F_1
- (c) F_1 is only a little greater than F_2
- (d) F_1 and F_2 are equal.

7. Mass of a man is 75 kg and he has reached the centre of the Earth. The mass and weight of man at the Earth's centre, with radius R , respectively, will be:

- (a) 75 kg, 75 N
- (b) 0 kg, 0 N
- (c) 75 kg, 0 N
- (d) 75 kg, 735 N

8. Rita throws the ball vertically upward with velocity u . The ball reaches the greatest height h to which it will rise before falling back, is given by:

- (a) u/g
- (b) $u^2/2g$
- (c) u^2/g
- (d) $u/2g$

9. An object weighs 10 N in air. When immersed fully in water, it weighs only 8 N. The weight of the liquid displaced by the object will be

- (a) 2 N
- (b) 8 N
- (c) 10 N
- (d) 12 N

10. Which of the following statements is true about the gravitational force?

- (a) It is always repulsive.
- (b) It acts only between planets.
- (c) It is the weakest force in nature but acts over large distances.
- (d) It depends on temperature.

11. If the mass of one body is doubled, the gravitational force between two bodies:

- (a) becomes half
- (b) becomes double
- (c) remains the same
- (d) becomes four times

12. Universal gravitational constant (G) has the SI unit:

- (a) $\text{N}\cdot\text{m}^2/\text{kg}^2$
- (b) m/s^2
- (c) N/kg
- (d) $\text{kg}\cdot\text{m}/\text{s}^2$

13. The force of gravitation between two objects does not depend on:

- (a) Their masses
- (b) Distance between them
- (c) Their temperatures
- (d) Universal gravitational constant

14. The weight of a body is:

- (a) The same everywhere
- (b) Maximum at the equator
- (c) Minimum at the poles
- (d) Varies with location due to gravity

15. The value of acceleration due to gravity (g) on the surface of the Earth is approximately:

- (a) $6.67 \times 10^{-11} \text{ m}/\text{s}^2$
- (b) $9.8 \text{ m}/\text{s}^2$
- (c) $98 \text{ m}/\text{s}^2$
- (d) $10 \text{ m}/\text{s}^2$

 (yaha Marks katate h)

Assertion And Reasons Questions

These consist of two statements - Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and Rare true, and (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true.

16. Assertion (A): G is a scalar quantity.

Reason (R): The value of G is constant throughout the universe and is unaffected by direction.

17. Assertion (A): g is different at poles and equator.

Reason (R): Earth is not a perfect sphere.

18. Assertion (A): A small object experiences a small buoyant force.

Reason (R): Buoyant force depends on volume and density of fluid.

19. Assertion (A): A bucket of water feels heavier when taken out of water.

Reason (R): A buoyant force acts on the bucket inside the water which reduces its weight.

20. Assertion (A): At the center of the Earth, a body has no centre of gravity.

Reason (R): $g = 0$ at the centre of Earth.

 (pahle points socho firr likho)

SHORT ANSWER TYPE QUESTIONS (2 - 3 Marks)

21. On the Earth, a stone is thrown from a height in a direction parallel to the Earth's surface while another stone is simultaneously dropped from the same height. Which stone would reach the ground first and why?

22. What is the direction of action of the buoyant force on an item immersed in liquid?

23. A ball falls from 20 m height. Calculate its velocity before hitting ground.

24. Why do astronauts feel weightless in space?

25. Which phase will exert more pressure on ground, when a man is standing or

26. Why is the weight of an object on the Moon $\frac{1}{6}$ th its weight on Earth?

27. A wooden block floats in water with $\frac{3}{4}$ of its volume submerged. What is its density?

28. (A) When a body is thrown vertically upwards, that will be its velocity as maximum height?
(B) What is more fundamental: mass or weight?
29. (A) Does the force of gravitation depend on nature of medium that separate the two objects?
(B) When does the gravitational force of attraction become really large?
30. A piece of copper of mass 120 g is dipped in a measuring cylinder containing water at 30 mL mark. The water rises to 42 mL mark. Find
(A) volume of the copper piece.
(B) the density of the copper.
31. If the volume of 50 g of a substance is 20 cm^3 and the density of water is 1 g/cm^3 , will the substance float or sink? Why?
32. State how the value of g changes with depth and altitude.
33. If the Earth's density is halved but its radius remains unchanged, what will be the change in acceleration due to gravity?

 (pahle points socho firr likho)

Long Answer Type Questions (5 Marks)

34. . Two objects Also, of masses m_1 and m_2 having the same size are dropped simultaneously from heights h_1 and h_2 respectively. Find out the ratio of time they would take in reaching the ground. Will this ratio remain the same if:
(A) one of the objects is hollow and the other one is solid and
(B) both of them are hollow, size remaining the same in each case ? give reason.
35. . A man is standing at the top of a 60 m high tower. He throws a ball vertically upwards with a velocity of 20 m/s. After what time, will the ball pass him going downwards? How long after its release, will the ball reach the ground?
36. . A ball is permitted to fall from the top of a 200-metre-high tower while another ball is projected vertically upwards from the round at a speed

of 50 metres per second. Determine when and where the two stones will meet .

37. (a) State Archimedes' Principle.

(b) Why does an iron nail sink in water but a ship made of iron float?

(c) A body weighs 500 g in air and 300 g in water. Find the buoyant force acting on it.

38. (a) Define density and relative density.

(b) Why is the relative density of mercury 13.6?

(c) A solid weighs 100 g in air and 80 g in water. Find its relative density.

39. Two objects of masses 10 kg and 20 kg are placed 2 m apart. Calculate the gravitational force between them. (Take $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$)

40. A body of volume 0.05 m^3 is fully immersed in water. Calculate the buoyant force acting on it. (Density of water = 1000 kg/m^3 , $g = 9.8 \text{ m/s}^2$)

41. The value of acceleration due to gravity on the surface of the Moon is about $1/6$ th that on Earth. Explain how the weight of an object changes on the Moon. Also derive the relationship between weight and mass. If a person weighs 60 N on Earth, what would be his weight on the Moon? Why does his mass remain unchanged?

42. Define free fall and acceleration due to gravity.

An object is dropped from a height of 78.4 m.

(a) How long will it take to reach the ground?

(b) With what velocity will it strike the ground?

(read paragraph first)

Case Study/Source Based Question

43. Sahil was driving the car at a very high speed to reach the hospital and he tried to drift the car. But the car slides from a ledge and falls to the ground in 2 seconds due to brake failure. Take acceleration due to gravity, $g = 10 \text{ m/s}^2$.

(A) What is the speed at which the car hits the ground?

(B) What is the height of the ledge above the ground?

(C) What is more fundamental: mass or weight?

44. There are many forces at work in the universe, as well as many pushes and pulls. Even if it's just the ground, we're always pushing or pulling something. However, it turns out that there are just four fundamental forces in Physics, from which everything else is derived: The strong, weak, electromagnetic, and gravitational forces. The gravitational force is a force that attracts all mass-bearing objects. The gravitational force is referred to as attractive because it always strives to pull masses together rather than pushing them apart. In reality, everything in the cosmos, including you, is tugging on every other object! This is referred to as Newton's Universal Law of Gravitation.

(A) On the surface of the Earth, a body of mass 1 kg is attracted by the Earth with a force equal to:

- (a) 1 N
- (b) 98 N
- (c) 9.8 N
- (d) 10 N.

(B) What is the mass of an object whose weight is 98 N?

- (a) 10 kg
- (b) 1 kg
- (c) 1 g
- (d) 10 g

(C) The gravitational force causes:

- (a) Tides
- (b) Motion of moon
- (c) None of them
- (d) Both (a) and (b)

(D) Assertion: The Earth and the ball attract each other with an equal gravitational force. Reason: The Earth moves towards the ball.

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and R are true, and (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true.

(E) The SI unit of upthrust is:

- (a) Newton
- (b) m/s
- (c) Pascal
- (d) m/s^2

PK Special [KBC]



45. Why does the value of 'g' decrease as we go above the Earth's surface or inside the Earth? Explain

46. Explain why a person feels lighter while swimming. Name the force responsible for this.

47. Why do objects weigh less on the Moon than on Earth? Give numerical comparison.

**48. Calculate the gravitational force between two masses of 5 kg and 8 kg placed 0.6 m apart.
(Take $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$)**

49. A 2 kg object is dropped from a height of 20 m. Calculate:

(a) Potential energy at the top

(b) Speed just before hitting the ground

($g = 9.8 \text{ m/s}^2$)

SOLUTIONS

MULTIPLE CHOICE TYPE QUESTIONS

1. A
2. D
3. C
4. A
5. C
6. D
7. C
8. B
9. A
10. C
11. B
12. A
13. C
14. D
15. B

ASSERTION AND REASONS TYPE QUESTIONS:

16. (a)
17. (a)
18. (a)
19. (a)
20. (a)

SHORT ANSWER TYPE QUESTIONS

21. Both stones will reach the ground at the same time.

Reason:

- The vertical motion of both stones is influenced only by the acceleration due to gravity (g).
- The stone thrown horizontally has an extra horizontal velocity, but this does not affect its vertical fall.
- The time taken to fall from a certain height depends only on vertical motion and initial vertical velocity (which is zero for both).

Thus, since both are falling under gravity from the same height and have zero initial vertical velocity, they will take the same time to reach the ground

SOLUTIONS

22. The buoyant force always acts in the upward direction, opposite to the direction of gravity.

- When an object is immersed in a fluid, the pressure at the bottom of the object is greater than at the top due to fluid depth.
- This pressure difference causes a net upward force called the buoyant force.
- Hence, it acts vertically upward through the center of buoyancy.

23. Given:

Height (h) = 20 m

Initial velocity (u) = 0 (free fall)

Acceleration due to gravity (g) = 9.8 m/s^2

We use the formula:

$$v^2 = u^2 + 2gh$$

$$v^2 = 0 + 2 \times 9.8 \times 20 = 392$$

$$v = \sqrt{392} = 19.8 \text{ m/s (approx.)}$$

24. Astronauts feel weightless in space because they are in a state of free fall inside their spacecraft.

Both the spacecraft and astronauts fall toward the Earth due to gravity with the same acceleration.

- As there is no normal reaction force acting on them, they experience a condition of zero apparent weight.
- This sensation is called weightlessness, though gravitational force still acts on them.

25. Standing will exert more pressure because pressure = force / area.

When standing, the area of contact is less, so pressure is more.

26. Because the acceleration due to gravity (g) on the Moon is 1/6th that on Earth.

Since weight = mass $\times$ g, the weight becomes 1/6th.

27. The fraction of volume submerged = density of block / density of water

Given:

Volume submerged = $3/4$

So,

Density of block = $3/4 \times 1000 = 750 \text{ kg/m}^3$

SOLUTIONS

28. (A) When a body is thrown vertically upwards, its velocity decreases due to gravity.

At the maximum height, the velocity becomes zero before the body starts falling back down.

(B) Mass is more fundamental than weight.

- Mass is a scalar quantity and remains constant everywhere.
- Weight = mass $\times$ gravity ($W = mg$) and depends on gravity, which varies with location.

29. (A) No, the gravitational force between two objects does not depend on the medium between them.

It depends only on their masses and distance between them, as per Newton's law of gravitation.

(B) The gravitational force becomes large when:

- The masses involved are very large (like Earth and Moon).
- The distance between the objects is very small (since force is inversely proportional to square of distance).

30. (A) Initial water level = 30 mL, Final level = 42 mL

Volume of copper piece = $42 - 30 = 12 \text{ cm}^3$

(B) Mass = 120 g, Volume = 12 cm^3

Density = Mass / Volume = $120 / 12 = 10 \text{ g/cm}^3$

31. Mass = 50 g, Volume = 20 cm^3

Density of substance = $50 / 20 = 2.5 \text{ g/cm}^3$

Density of water = 1 g/cm^3

Since the substance is denser than water, it will sink in water.

32. The acceleration due to gravity (g) changes with both depth and altitude:

- As we go deeper into the Earth, value of g decreases because the effective mass attracting the object becomes less.
- As we go to a higher altitude, g also decreases because the distance from Earth's center increases, reducing gravitational pull.

33. We know:

$g \propto \text{density of Earth}$ ($g = \frac{4}{3} \times \pi \times R \times \text{density}$)

If the density is halved but radius remains the same, then:

New $g = \frac{1}{2} \times \text{original } g$

So, acceleration due to gravity will become half its original value.

SOLUTIONS

34. The time taken by an object falling freely from height h is given by:

$$t = \sqrt{2h / g}$$

So, for two objects dropped from heights h_1 and h_2 , their time of fall will be:

$$t_1 = \sqrt{2h_1 / g}$$

$$t_2 = \sqrt{2h_2 / g}$$

Therefore, the ratio of time taken:

$$t_1 / t_2 = \sqrt{h_1 / h_2}$$

This ratio depends only on the heights from which they are dropped, not on mass or size.

(A) One object is hollow, other solid:

In vacuum, both fall with the same acceleration due to gravity. So, the time taken depends only on height — ratio remains same.

In air, a hollow object may experience more air resistance, so it may fall slightly slower. Hence, the ratio may vary slightly.

(B) Both are hollow, same size:

Since both are affected equally by air resistance, the time taken depends only on their heights. The ratio remains same. $t_1 / t_2 = \sqrt{h_1 / h_2}$

Ratio remains the same in vacuum. In air, it can vary if air resistance differs.

35. Given:

- Initial velocity, $u = 20 \text{ m/s}$
- Acceleration due to gravity, $g = 9.8 \text{ m/s}^2$
- Height of tower = 60 m

(A) Time to reach back to same height (where man is standing):

At maximum height, final velocity $v = 0$

Using the equation:

$$v = u - gt$$

$$0 = 20 - 9.8t$$

$$\Rightarrow t = 20 / 9.8 = 2.04 \text{ s}$$

So, the ball will pass the man after 2.04 seconds while coming down.

(B) Time to reach the ground:

First, find maximum height reached by the ball:

SOLUTIONS

35. $h = u^2 / (2g) = 20^2 / (2 \times 9.8) = 400 / 19.6 = 20.41 \text{ m}$

Total height from maximum point to ground:

$H = 60 + 20.41 = 80.41 \text{ m}$

Now, time taken to fall from max height to ground:

$s = (1/2)gt^2 \Rightarrow t = \sqrt{(2s / g)}$

$t = \sqrt{(2 \times 80.41 / 9.8)} = \sqrt{(16.41)} \approx 4.05 \text{ s}$

Total time = time to reach top + time to fall = $2.04 + 4.05 = 6.09 \text{ seconds}$

36. Let the two balls meet after t seconds.

Let:

- Height of the tower = 200 m
- For Ball A (falling from the top):
- Initial velocity, $u_1 = 0 \text{ m/s}$,
- Distance covered in time $t = s_1 = \frac{1}{2} \cdot g \cdot t^2$
- For Ball B (rising from ground):
- Initial velocity, $u_2 = 50 \text{ m/s}$,
- Distance covered = $s_2 = u \cdot t - \frac{1}{2} \cdot g \cdot t^2$

They meet when the sum of distances covered by both = total height of the tower:

So,

$s_1 + s_2 = 200$

$\Rightarrow \frac{1}{2} \cdot g \cdot t^2 + (50 \cdot t - \frac{1}{2} \cdot g \cdot t^2) = 200$

$\Rightarrow 50 \cdot t = 200$

$\Rightarrow t = 4 \text{ seconds}$

We can use the distance travelled by the second ball (rising one):

$s = u \cdot t - \frac{1}{2} \cdot g \cdot t^2$

$= 50 \times 4 - \frac{1}{2} \times 9.8 \times 4^2$

$= 200 - 0.5 \times 9.8 \times 16$

$= 200 - 78.4$

$= 121.6 \text{ m}$

- The two balls will meet after 4 seconds
- They will meet at a height of 121.6 metres above the ground

37. Let the height of the tower = 200 m

Ball A is dropped from the top (initial velocity $u_1 = 0$)

Ball B is projected upwards from the ground with $u_2 = 50 \text{ m/s}$

Let they meet after t seconds

- For Ball A (falling):
- $s_1 = \frac{1}{2} \cdot g \cdot t^2 = \frac{1}{2} \times 9.8 \times t^2$

SOLUTIONS

For Ball B (rising):

$$s_2 = u \cdot t - \frac{1}{2} \cdot g \cdot t^2 = 50 \cdot t - \frac{1}{2} \times 9.8 \times t^2$$

Now,

$$s_1 + s_2 = 200$$

$$\Rightarrow (\frac{1}{2} \cdot g \cdot t^2) + (50 \cdot t - \frac{1}{2} \cdot g \cdot t^2) = 200$$

$$\Rightarrow 50 \cdot t = 200$$

$$\Rightarrow t = 4 \text{ seconds}$$

Now, distance from ground = $s_2 = 50 \cdot t - \frac{1}{2} \cdot 9.8 \cdot t^2$

$$= 50 \times 4 - \frac{1}{2} \times 9.8 \times 16$$

$$= 200 - 78.4$$

$$= 121.6 \text{ m}$$

The balls meet after 4 seconds at a height of 121.6 m above the ground.

38. (a) Archimedes' Principle:

When a body is immersed in a fluid, it experiences an upward force (buoyant force) equal to the weight of the fluid displaced by it.

(b) An iron nail sinks because it displaces less water than its own weight, so the buoyant force is less.

A ship made of iron floats because it displaces a large volume of water, making the buoyant force equal to its weight.

(c)

$$\text{Buoyant Force} = \text{Weight in air} - \text{Weight in water}$$

$$= 500 \text{ g} - 300 \text{ g} = 200 \text{ g} = 2 \text{ N}$$

$$(1 \text{ g} = 0.01 \text{ N})$$

39. Formula:

$$F = G \cdot m_1 \cdot m_2 / r^2$$

Given:

$$m_1 = 10 \text{ kg}, m_2 = 20 \text{ kg}, r = 2 \text{ m}$$

$$G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$$

$$F = (6.67 \times 10^{-11} \times 10 \times 20) / 2^2$$

$$= (1334 \times 10^{-11}) / 4$$

$$= 3.335 \times 10^{-9} \text{ N}$$

40. Formula:

$$F_b = \rho \cdot V \cdot g$$

SOLUTIONS

Given: $\rho = 1000 \text{ kg/m}^3$, $V = 0.05 \text{ m}^3$, $g = 9.8 \text{ m/s}^2$

$$F_b = 1000 \times 0.05 \times 9.8$$

$$= 490 \text{ N}$$

41. Weight = mass $\times$ acceleration due to gravity

On Moon, $g_{\text{moon}} = (1/6) \cdot g_{\text{earth}}$

$$W_{\text{moon}} = m \times g/6 = W_{\text{earth}} / 6$$

If $W_{\text{earth}} = 60 \text{ N}$, then:

$$W_{\text{moon}} = 60 / 6 = 10 \text{ N}$$

Mass remains unchanged because it is independent of gravity

42. Free fall: Motion of a body under gravitational force alone is called free fall.

Acceleration due to gravity (g): Acceleration of a body in free fall. On Earth, $g = 9.8 \text{ m/s}^2$

Given: $h = 78.4 \text{ m}$, $g = 9.8 \text{ m/s}^2$

(a)

$$t = \sqrt{(2 \cdot h / g)}$$

$$= \sqrt{(2 \times 78.4 / 9.8)}$$

$$= \sqrt{16} = 4 \text{ s}$$

(b)

$$v = g \cdot t = 9.8 \times 4 = 39.2 \text{ m/s}$$

CASE STUDY/SOURCE BASED QUESTION:

43. A. We use the formula:

$$v = u + g \cdot t$$

where:

- $u = 0 \text{ m/s}$ (initial velocity, as the car starts falling from rest),
- $g = 10 \text{ m/s}^2$ (acceleration due to gravity),
- $t = 2 \text{ s}$

So,

$$v = 0 + 10 \times 2 = 20 \text{ m/s}$$

B. Use the formula:

$$s = ut + (1/2) \cdot g \cdot t^2$$

Here,

SOLUTIONS

- $u = 0 \text{ m/s}$,
- $g = 10 \text{ m/s}^2$,
- $t = 2 \text{ s}$

So,

$$s = 0 + \frac{1}{2} \times 10 \times (2)^2 = 5 \times 4 = 20 \text{ m}$$

C. Mass is a fundamental quantity because it is constant everywhere in the universe and does not change with gravity.

Weight depends on gravitational force and varies from place to place (e.g., on Moon or Earth).

Mass is more fundamental than weight because it is a constant physical quantity, whereas weight depends on gravity.

44. (A) $\rightarrow$ (c)

(B) $\rightarrow$ (a)

(C) $\rightarrow$ (d)

(D) $\rightarrow$ (c)

(E) $\rightarrow$ (a)

PK Special [KBC]

45. The value of 'g' (acceleration due to gravity) decreases as we go above the Earth's surface or inside the Earth because:

- Above Earth's Surface: As height increases, the distance from the center of the Earth increases. Since $g \propto 1/R^2$, the gravitational pull weakens with distance.
- Inside the Earth: As we go deeper, the mass of the Earth above us no longer contributes to gravity, so the effective mass attracting the object decreases. Hence, g decreases.

46. A person feels lighter while swimming because of the buoyant force exerted by water.

This force acts upward and opposes the weight of the body, reducing the net downward force, making the person feel lighter.

Force responsible: Buoyant force (Upthrust)

47. On the Moon, objects weigh less because the gravitational force is weaker.

$$g_{\text{moon}} = \frac{1}{6} \times g_{\text{earth}}$$

If a body weighs 60 N on Earth:

SOLUTIONS

Weight on Moon = $60 \div 6 = 10 \text{ N}$

Hence, objects weigh 1/6th on the Moon compared to Earth.

48. Given:

$$m_1 = 5 \text{ kg}, m_2 = 8 \text{ kg}, r = 0.6 \text{ m},$$

$$G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$$

Gravitational Force (F)

$$F = G \times (m_1 \times m_2) / r^2$$

$$F = (6.67 \times 10^{-11}) \times (5 \times 8) / (0.6)^2$$

$$F = (6.67 \times 10^{-11} \times 40) / 0.36$$

$$F = 2.96 \times 10^{-9} \text{ N}$$

Answer: $2.96 \times 10^{-9} \text{ N}$

49. Given:

$$m = 2 \text{ kg}, h = 20 \text{ m}, g = 9.8 \text{ m/s}^2$$

(a) Potential Energy (PE):

$$PE = mgh = 2 \times 9.8 \times 20 = 392 \text{ J}$$

(b) Speed just before hitting ground:

$$\text{Using: } v^2 = 2gh$$

$$v^2 = 2 \times 9.8 \times 20 = 392$$

$$v = \sqrt{392} = 19.8 \text{ m/s (approx)}$$

SOLUTIONS

49. Voluntary muscles (also called skeletal or striated muscles) are long, cylindrical, multinucleated, and striated. These muscles are arranged in bundles and are connected to bones. Their structure allows for quick and powerful contraction, which is under conscious control. This helps in activities like walking, writing, and lifting objects.

Involuntary muscles, such as smooth muscles and cardiac muscles, are spindle-shaped (smooth) or branched (cardiac) and have one nucleus. These muscles contract slowly and rhythmically without conscious control. They are found in internal organs like the stomach, intestines (smooth), and the heart (cardiac).

Comparison:

Voluntary muscles → Rapid, controlled movement (due to striated, bundled structure)

Involuntary muscles → Slow, automatic functions (due to smooth or branched structure)

Thus, the structure of muscle fibers is directly related to their function in body movement and control.

50. Xylem is a complex permanent tissue in plants responsible for the transport of water and minerals from the roots to other parts of the plant. It also provides mechanical strength.

Xylem Components:

- 1. Tracheids – Long, narrow dead cells with thick lignified walls and tapering ends.**
- 2. Vessels – Wider, tube-like dead cells joined end-to-end to form long pipelines.**
- 3. Xylem Fibres – Dead, thick-walled supportive cells.**
- 4. Xylem Parenchyma – Living cells that store food and help in lateral conduction.**

Function:

- Conducts water and mineral salts from roots to shoots.**
- Provides structural support to the plant.**

Importance of Lignin Deposition:

Lignin is a complex organic substance that is deposited in the cell walls of tracheids and vessels. It makes the walls thick, rigid, and waterproof, which prevents them from collapsing under the pressure of water transport. It also helps maintain the integrity and strength of xylem tissues.

51. Neuron is the basic structural and functional unit of the nervous system. It is specialised to transmit nerve impulses quickly and efficiently across long distances.

Structure of a Neuron:

- 5. Dendrites – Short, branched projections that receive signals.**
- 6. Cell Body (Cyton) – Contains the nucleus and cytoplasm; integrates signals.**

SOLUTIONS

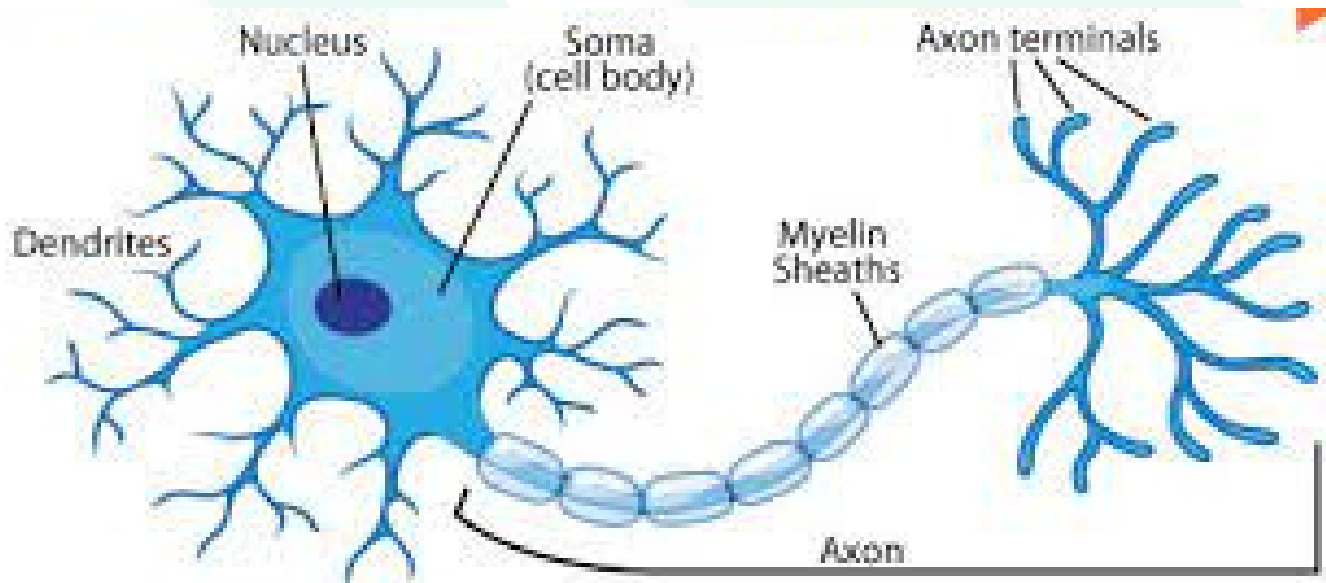
1. **Axon** – A long, tubular extension that carries impulses away from the cell body.
2. **Axon Terminals** – Pass signals to other neurons or effectors.
3. **Myelin Sheath** – Fatty layer covering the axon, increases impulse speed.

Function:

Neurons receive, process, and transmit electrical impulses to other neurons, muscles, or glands.

Difference from Muscle Tissues:

- Neurons transmit electrical signals; muscle tissues contract to produce movement.
- Neurons have branched and long extensions (axon, dendrites); muscle cells are elongated and fibrous.
- Neuron signals are electrical and chemical, while muscles work via mechanical contraction.



52. Desert plants (xerophytes) are adapted to conserve water in harsh, dry environments. They have:

- **Thick cuticle:** A waxy layer on the epidermis that prevents excessive water loss through evaporation.
- **Sunken stomata:** Stomata are located in pits to trap moisture and reduce transpiration.

Tissues Involved:

- **Epidermal tissue** (protective tissue) forms the outer covering with cuticle.
- **Parenchyma tissue** in leaves stores water and may assist in photosynthesis.

These adaptations help desert plants survive in conditions where water is scarce.